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Multi-objective optimization of a rectangular micro-channel heat sink using the augmented ϵ-constraint method

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ABSTRACT

This article proposes the use of an improved version of the ε-constraint method for the design and optimization of a rectangular micro-channel heat sink. This study aims to optimize the geometric parameters describing the micro-channel, namely its width, its fin width and its depth. The thermal resistance and the pumping power, considered as indicators of the thermal performance, have been approximated by surrogated models based on response surface approximation. The optimization task is formulated as a nonlinear programming problem. This approach has been implemented in GAMS. Sufficient details on both the single-criterion and multi-criteria formulation of the problem have been provided. The implications of the mathematical modelling formulation and the interrelationship between criteria and estimated quantities have been clarified. The use of the augmented ϵ-constraint method for the multi-objective optimization of a rectangular micro-channel heat sink constitutes the main contribution of this work.

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Micro-channel; heat sink; optimization; ε-constraint; AUGMENCON

Nomenclature

eps	A very small number
h	Specific enthalpy (J/kg)
H_c	Micro-channel depth (m)
k	Thermal conductivity (W/m K)
L_x	Length of the heat sink (m)
L_y	Width of the heat sink (m)
L_z	Height of the heat sink (m)
min	Minimize
max	Maximize
P	Pumping power (W)
q	Heat flux (W/m^2)
R_{TH}	Thermal resistance (K/W)
r_k	Range of kth objective function
s_k	Surplus of ith objective
T	Temperature (K)
u	Liquid velocity (m/s)
W_c	Micro-channel width (m)
w_k	Objective function component weight

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W_w Fin width (m)
x,y,z Orthogonal coordinate system

Greek symbols

θ Design variables w_c/h_c
ϕ Design variables w_w/h_c
ξ Objective function
ε_k Right-hand side of kth objective function
τ_{ij} Stress tensor

Subscripts

f Fluid
i Inlet
o Outlet
s Substrate

1. Introduction

Recent advances in the development of the abilities of electronic devices have brought a series of challenges related to the management of their temperature level. Various cooling schemes have been proposed to address this challenge, of which micro-channel heat sinks are described as one of the most successful and effective methods (Adham, Mohd-Ghazali, and Ahmad 2015) (Figure 1). To enhance the cooling capabilities of micro-channel heat sinks, different geometries are used, such as circular (Hetsroni *et al.* 2004), rectangular (Husain and Kim 2010), triangular (Tiselj *et al.* 2004), trapezoidal (McHale and Garimella 2010) and hexagonal micro-channels (Perret, Schaeffer, and Boussey 1998). A study conducted by Hetsroni *et al.* (2005) reveals that rectangular micro-channel heat sinks produce better performance and guarantee more stability compared to other geometric configurations of channels.

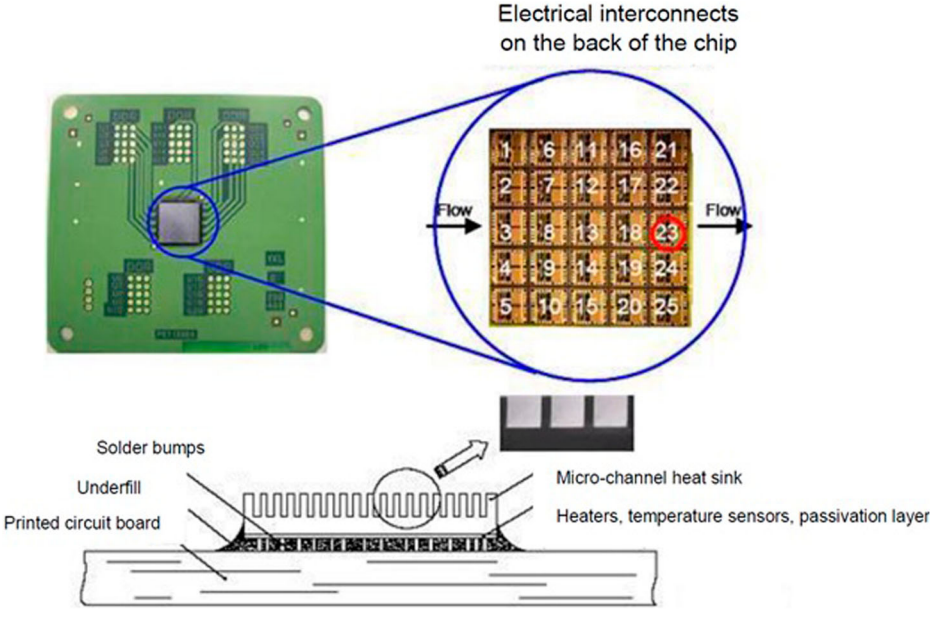


Figure 1. Micro-channel heat sink mounted on a printed circuit board (As described by Harirchian and Garimella 2009).

A study on the development and optimization of a micro-channel heat sink, conducted by Ryu, Choi, and Kim (2003), reveals that the channel width and depth appear to be more critical than other factors with respect to heat sink performance. The optimum design parameters of micro-channel heat sinks, namely the channel number and fin thickness, improve its performance, as suggested by the numerical study conducted by Jo *et al.* (2001). To optimize the geometric structure of a rectangular micro-channel heat sink, a detailed numerical simulation of the heat transfer was performed by Li and Peterson (2006). This study, on silicon–water micro-heat sinks, suggests that the optimal configuration for rectangular channel heat sinks is expected when the number of channels approaches 120 channels per centimetre. The effects of mini-channel dimensions, wall thickness and inlet velocity were investigated numerically by Xie *et al.* (2009). The thermal resistance and maximum allowable heat flux were used as indicators of performance. This study points out that narrow and deep channels, having relatively thin bottom and wall thicknesses, improved the heat transfer performance.

Several advanced optimization approaches have been considered in the past decade. For example, numerical optimization, with the help of surrogate analysis, was performed by Husain and Kim (2009). A single-objective optimization problem with thermal resistance considered as the objective function and pumping power as a constraint was developed. This work provided clarity on the economic compromise with design variables with respect to the fabrication methods and the available pumping power sources. In other studies by Husain and Kim (2008a, 2010), a hybrid multi-objective evolutionary algorithm (MOEA) consisting of a surrogate analysis and an evolutionary algorithm was used to optimize the geometry of trapezoidal micro-channels. With thermal resistance and pumping power considered as objectives functions, global Pareto-optimal solutions describing the best geometric configurations were obtained. It appears that most of these studies used a combination of numerical simulation and surrogate methods in the formulation of thermal resistance and pumping power considered as objective functions. Hence, the use of these methods requires detailed knowledge of their theoretical development. In addition, the complexities of the nature of these methods and the computational time consumed have led to a search for different approaches for optimization.

2. Motivation

Various optimization methods, other than the MOEA mentioned in Section 1, have been applied by previous researchers for dimensional optimization of micro-channel heat sinks:

- Hu and Xu (2009) proposed the use of the sequential quadratic programming (SQP) method for the design optimization of the structure size of a micro-channel.
- A genetic algorithm (GA) was used for the design optimization of a micro-heat sink for concentrating photovoltaic/thermal systems by Karathanassis *et al.* (2013).
- A non-dominated sorting genetic algorithm (NSGA2) was used by Husain and Kim (2008) to perform the optimization of micro-channel heat sinks.
- The improved strength Pareto evolutionary algorithm (SPEA2) was considered by Adham, Mohd-Ghazali, and Ahmad (2015) for the optimization of a rectangular micro-channel heat sink.
- A Jaya algorithm and a teaching–learning-based optimization (TLBO) algorithm were applied for the dimensional optimization of micro-channel heat sink by Rao *et al.* (2016).
- A multi-parameter optimization approach, integrating the simplified conjugate gradient method and a full three-dimensional heat transfer and flow model, was used by Wang *et al.* (2011) to optimize the geometry of micro-channel heat sinks.
- A three-dimensional fluid–solid conjugated micro-channel heat sink model that combined the simplified conjugate-gradient method was used as the optimization tool in a study by Wang *et al.* (2013).
- A combined optimization scheme that incorporated a simplified conjugate-gradient method and a three-dimensional nanofluid-cooled micro-channel heat sink model was proposed by Wang, An,

and Xu (2013) to obtain the optimal geometric structure for a silicon-based micro-channel heat sink.

- Leng *et al.* (2015a) applied a multi-parameter optimization algorithm for the optimization of a double-layered micro-channel heat sink with truncated top channels to enhance its cooling performance.
- Single- and multi-objective optimization approaches were performed by Leng *et al.* (2015b) to improve the temperature uniformity on the bottom wall and to reduce the overall thermal resistance for double-layered micro-channel heat sinks considering constant pumping powers.

This is by no means a comprehensive list of the 'dimensional optimization' of a micro-channel heat sink, but it serves as a thorough overview of studies conducted by numerous researchers. The present study focuses on the optimization of a rectangular micro-channel heat sink. In this work, an improved version of the augmented ε-constraint method (AUGMENCON2) is proposed to study the performance of heat transfer and fluid flow in the micro-channel heat sink by considering thermal resistance and pumping power as objective functions. The proposed AUGMENCON2 method was developed by Mavrotas and Florios (2013). A nonlinear programming (NLP) problem is formulated and implemented in a high-level modelling system for mathematical programming and optimization, called the General Algebraic Modeling System (GAMS). This new multi-objective approach is meant to produce exact Pareto sets, corresponding to the best geometric configuration of a micro-channel, with a reasonable level of accuracy and short computational time.

The remainder of this article is organized as follows. The model considered in this study is presented in Section 3. Section 4 presents the fundamental parameters and equations considered in the modelling of the micro-channel heat sink. In Section 5, a discussion on single-objective optimization is summarized. Section 6 considers the optimization of multiple objectives using AUGMENCON2. Section 7 reports the results and implications of this work, and conclusions are presented in Section 8.

3. Description of the problem

In this section, a 10 mm × 10 mm × 0.5 mm model (as depicted in Figure 2) is considered for thermal performance analysis and optimization using AUGMENCON2. Water is considered as a coolant flowing into the micro-channel and leaving at the outlet. The remaining portion of the heat sink is occupied by silicon substrate. The latter is maintained as an adiabatic condition. No-slip conditions ($u = 0$) are assumed at the boundary of the inner part of the micro-channel. In this study, steady, incompressible, laminar and conjugate heat transfer conditions are considered with constant fluid (water) properties. In addition, radiation heat transfer is assumed to be negligible. Therefore, the heat transfer in the micro-channel, describing the conservation of mass, momentum and energy, can be written as follows (Husain and Kim 2008b; Fedorov and Viskanta 2000):

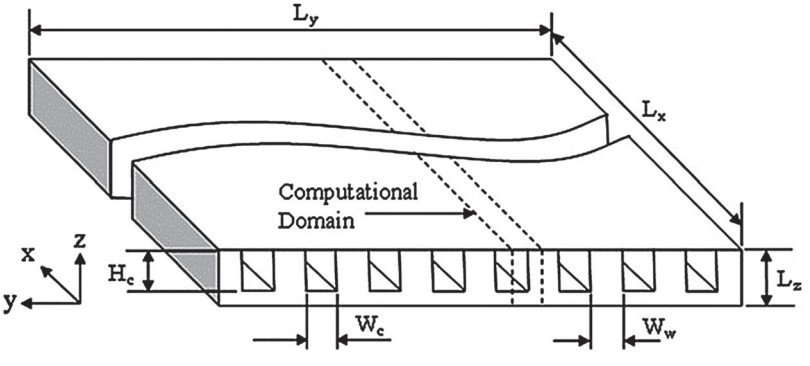


Figure 2. Computational domain of a micro-channel heat sink.

Mass:

$$\frac{\partial \rho_f}{\partial t} + \frac{\partial (\rho_f u_i)}{\partial x_i} = 0 \tag{1}$$

Momentum:

$$\rho_f \frac{\partial u_i}{\partial t} + \rho_f u_j \frac{\partial u_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \rho_f g_i + \frac{\partial}{\partial x_j}\left(u_f \frac{\partial u_i}{\partial x_j}\right) + \frac{\partial}{\partial x_j}\left(u_f \frac{\partial u_j}{\partial x_i}\right) - \frac{2}{3}\frac{\partial}{\partial x_i}\left(u_f \frac{\partial u_k}{\partial x_k}\right) \tag{2}$$

Energy (in fluid):

$$\rho_f \frac{\partial h}{\partial t} + \rho_f u_j \frac{\partial h}{\partial x_i} = \frac{\partial p}{\partial t} + u_i \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_i}\left(k_f \frac{\partial T_f}{\partial x_i}\right) + \tau_{ij}\frac{\partial u_i}{\partial x_j} \tag{3}$$

Energy (in solid):

$$\frac{\partial}{\partial x_i}\left(k_s \frac{\partial T_s}{\partial x_i}\right) = 0 \tag{4}$$

The following thermal boundary conditions in the z-direction have been considered:

$$-k_s\left(\frac{\partial T_s}{\partial x_i}\right) = q \text{ at } z = 0 \quad \text{and} \quad k_s\left(\frac{\partial T_s}{\partial x_i}\right) = 0 \text{ at } z = l_z \tag{5}$$

4. Design variables and objective functions

4.1. Geometric parameters

Two different variables (θ and ϕ) describing the micro-channel heat sink have been considered, $\theta = W_c/H_c$ and $\phi = W_w/H_c$, where w_c is the micro-channel width, W_w is the fin width, and H_c is the micro-channel depth. The latter has been kept at 400 μm during this entire optimization.

$$\begin{array}{l} \theta_{\min} < \theta < \theta_{\max} \\ \phi_{\min} < \phi < \phi_{\max} \\ \theta \text{ and } \phi \epsilon \Re^{+} \end{array} \tag{6}$$

4.2. Thermal resistance and pumping power

In this study, the thermal performance of the micro-channel heat sink is measured, with thermal resistance (R_{TH}) and pumping power (P). These indicators are considered as the two objective functions in the formulation of the mathematical programming problem. The thermal resistance and pumping power are obtained as follows (Husain and Kim 2008b):

$$R_{\mathrm{TH}} = \frac{T_{s,o} - T_{s,i}}{A_s \times q} \tag{7}$$

$$P = n \times u_{\mathrm{avg}} \times A_c \times \Delta P \tag{8}$$

where $T_{s,o} - T_{s,i}$ is the maximum temperature difference in the micro-channel heat sink, A_s is the area of the substrate subjected to the heat flux q, n indicates the number of micro-channels, u_{avg} is the average velocity, A_c is the cross-sectional area of the micro-channel, and ΔP is the pressure drop. To predict the optimal design points, three surrogate models, namely, response surface approximation (RSA), kriging and radial basis neural network, have been proposed by Husain and Kim (2008c). It is important to emphasize that surrogate predictions are closely related to the nature and suitability of the problem, which could result in different optimum points based on the surrogate model adopted.

The three surrogate models, together with the computational fluid dynamics (CFD) calculations, predicted almost the same objective function values for the optimization of a rectangular micro-channel heat sink (Husain and Kim 2008c). Most importantly, good agreement of thermal resistance between the computational and experimental results has been reported.

In this work, surrogate models based on the RSA constructed by Husain and Kim (2008b) were used to approximate the two objective functions. These surrogate models were developed to avoid experimental or numerical expenses and to save time. Detailed descriptions of the calculation of the objective functions using Navier–Stokes and heat condition equations at specific design points are available in Husain and Kim (2008b). In addition, clarity on the generation of response surface curves with the help of CFD calculations is provided. RSA, as a methodology, consists of fitting a polynomial function for discrete responses obtained from these numerical calculations. While noteworthy, this derivation is beyond the scope of this work. Therefore, AUGMENCON2 is applied to obtain Pareto-optimal solutions. The following two functional forms of the objective functions were obtained from numerical calculations (details available in Husain and Kim 2008b):

$$R_{\mathrm{TH}} = 0.0964 + 0.3124\theta - 0.7005\phi - 1.1122\theta\phi + 0.6044\theta^2 + 4.8528\phi^2 \tag{9}$$

$$P = 0.9925 - 9.3955\theta + 3.5575\phi - 14.9550\theta\phi + 22.9024\theta^2 - 0.0706\phi^2 \tag{10}$$

These equations suggest that the two design variables (θ and ϕ) have a significant effect on the thermal performance and pumping power of a micro-channel heat sink. The optimum values of the dimensions of the micro-channel (W_c, W_w and H_c) can be obtained by solving Equations (9) and (10). The incorporation of design and manufacturing constraints would ease the decision-maker's dilemma since the Pareto-optimal solutions could be spread over the entire range of variables.

5. Single-objective optimization

All the expressions involved in the mathematical problem formulation (MPF) have been described in Section 4. Together with the following equations, they represent an NLP problem:

$$(\mathrm{MPF}) \min_{\phi,\Phi} \xi = w_1(R_{\mathrm{TH}}) + w_2(P) \tag{11}$$

This mathematical model characterizes the essential elements of a rectangular micro-channel heat sink. In the following discussion, restricted cases of objectives are analysed and tendencies of geometric parameters are identified with respect to each objective function.

5.1. Emphasizing thermal resistance

All proposed NLP problems are solved by GAMS 23.8 using LINDOGLOBAL solver on a personal computer Pentium IV, 2.2 GHz with 4 GB RAM. The following constraints (upper and lower bounds) have been enforced on geometric parameters for the solver to carry out the search for optimal solutions in those ranges:

$$\begin{aligned} \theta.\mathrm{lo} = 0.10;\ \theta.\mathrm{up} = 0.25; \\ q.\mathrm{lo} = 0.04;\ \phi.\mathrm{up} = 0.10; \end{aligned} \tag{12}$$

Setting the objective function weight $w_2 = 0$, the problem reduces to Equation (9) and variable restrictions (12). Objective function (11) becomes:

$$(\mathrm{MPF}) \min_{q,\Phi} \xi_{R_{\mathrm{TH}}} = R_{\mathrm{TH}} \tag{13}$$

In this proposed approach, the range adopted (Equation 12) is similar to previous studies to enable validation of the results obtained with this new approach. In Table 1, the optimal solutions that minimize $\xi_{R_{\mathrm{TH}}}$ are represented by letters marked with an asterisk.

Table 1. Optimal solutions minimizing thermal resistance.

	θ^*	ϕ^*	R^*_{TH}	CPU time (s)
x^*	0.100	0.084	0.100	1.429

Table 2. Optimal solutions minimizing pumping power.

	θ^*	ϕ^*	P^*	CPU time (s)
x^*	0.218	0.040	0.045	0.628

Table 3. Tendency of parameters when optimizing individual components.

	R_{TH}	P
θ	$\downarrow$	$\uparrow \neq$
ϕ	$\uparrow$	$\downarrow \neq$

Physically, this optimal solution can be interpreted as follows:

- Minimize the ratio between the micro-channel width and the micro-channel depth.
- Significantly increase the ratio between the micro-channel fin width and the micro-channel depth.

The minimum thermal resistance is:

$$R_{\text{THmin}} = 0.100 \tag{14}$$

5.2. Emphasizing pumping power

Pumping power P is emphasized by setting the objective function weight $w_1 = 0$. The problem is simplified to Equation (10) and variable restrictions (12).

$$\text{(MPF)} \min_{\theta,\phi} \xi_P = P \tag{15}$$

In Table 2, the optimal solutions that minimize ξ_P are represented by letters marked with an asterisk.

Physically, this optimal solution can be interpreted as follows:

- Significantly increase the ratio between the micro-channel width and the micro-channel depth.
- Minimize the ratio between the micro-channel fin width and the micro-channel depth.

5.3. Single-objective optima: variable analysis

Table 3 summarizes the results of Subsections 5.1 and 5.2, highlighting the behaviour of the parameters. For these objectives, $\uparrow$ indicates an increasing tendency, $\downarrow$ indicates a decreasing tendency, and $\neq$ indicates conflicting tension between parameters.

The conflicting nature of objective functions when considered individually makes the multi-objective optimization approach suitable for this problem. Hence, the two objective components have been optimized simultaneously in Section 6.

6. Emphasizing all objective components

This work proposes an improved version of the augmented ε-constraint method (AUGMENCON2). Hence, a summary of the original version of the augmented ε-constraint method (AUGMENCON) is

presented. Details on the implementation of the ε-constraint method, as opposed to the augmented ε-constraint method, are available in Mavrotas (2009) and Tartibu, Sun, and Kaunda (2015a).

6.1. Augmented ϵ-constraint method (AUGMENCON)

The original AUGMENCON, proposed by Mavrotas (2009), has been successfully implemented in numerous studies related to thermal systems (Tartibu, Sun, and Kaunda 2013, 2015b). AUGMENCON is a variant of the classical ε-constraint method addressing many of its weaknesses: (1) AUGMENCON uses a lexicographic optimization method to calculate the ranges of the objective functions; (2) unlike the classical ε-constraint method, the efficiency of the returned solutions is proven; and (3) AUGMENCON significantly reduces the computational time through the use of several acceleration techniques. Detailed descriptions of AUGMENCON with a practical example are available in Tartibu, Sun, and Kaunda (2013, 2015b). AUGMENCOM can be formulated as follows:

$$\max\left(f_1(x) + \text{eps} \times \left(\frac{s_2}{r_2} + \frac{s_3}{r_3} + \ldots + \frac{s_p}{r_p}\right)\right) \tag{16}$$

subject to

$$f_2(x) - s_2 = \varepsilon_2; f_3(x) - s_3 = \varepsilon_3, \ldots; f_p(x) - s_p = \varepsilon_p$$

$$\mathrm{x} \in \mathrm{s} \quad \text{and} \quad \mathrm{s_i} \in \Re^+$$

where $\varepsilon_2, \varepsilon_3, \ldots, \varepsilon_p$ are the parameters for the right-hand side after a specific iteration drawn from the grid points of the objective functions 2, 3, …, p. The range r_2, $r_3, \ldots, r_p$ and the surplus variables s_2, $s_3, \ldots,$ s_p correspond to objective functions and constraints, respectively, while eps is a small number, usually between 10^{-3} and 10^{-6} (Mavrotas 2009). The flowchart of the AUGMENCON method for optimization of a series of objective functions is illustrated in Figure 3.

6.2. Improved version of the augmented ε-constraint method (AUGMENCON2)

Practically, AUGMENCON2 is applied as follows (Mavrotas and Florios 2013):

- The lexicographic optimization is performed on the primary objective and the rest of the objective functions to find any alternative optima (unlike AUGMENCON).
- The objective function range is calculated from the pay-off table.
- Ranges are divided into q_k equal intervals using $(q_k - 1)$ intermediate equidistant grid points. Total $(q_k + 1)$ grid points are obtained to vary the right-hand side (ε_k) of the kth objective function. Ultimately, the total number of runs can be evaluated as $(q_2 + 1) \times (q_3 + 1) \times \ldots \times (q_p + 1)$, where p is the number of competing objective functions.

In AUGMENCON2, the MPF defined by Equation (16) is modified slightly, as follows:

$$\max\left(f_1(x) + \text{eps} \times \left(\frac{s_2}{r_2} + 10^{-1}\frac{s_3}{r_3} + \ldots + 10^{-(p-2)}\frac{s_p}{r_p}\right)\right) \tag{17}$$

This modification is done with the intent of performing lexicographic optimization, not only on the primary objective function, but also on the rest of the objective functions to identify any alternative optima (Mavrotas and Florios 2013). Hence, the solver is forced to perform the sequential optimization of the constrained objectives f_1, f_2, f_3 to f_p and successively find their optima, unlike the previous formulation (Equation 16), where the sequence of optimization of f_2 to f_p was unimportant.

The flowchart of AUGMENCON2 is shown in Figure 4.

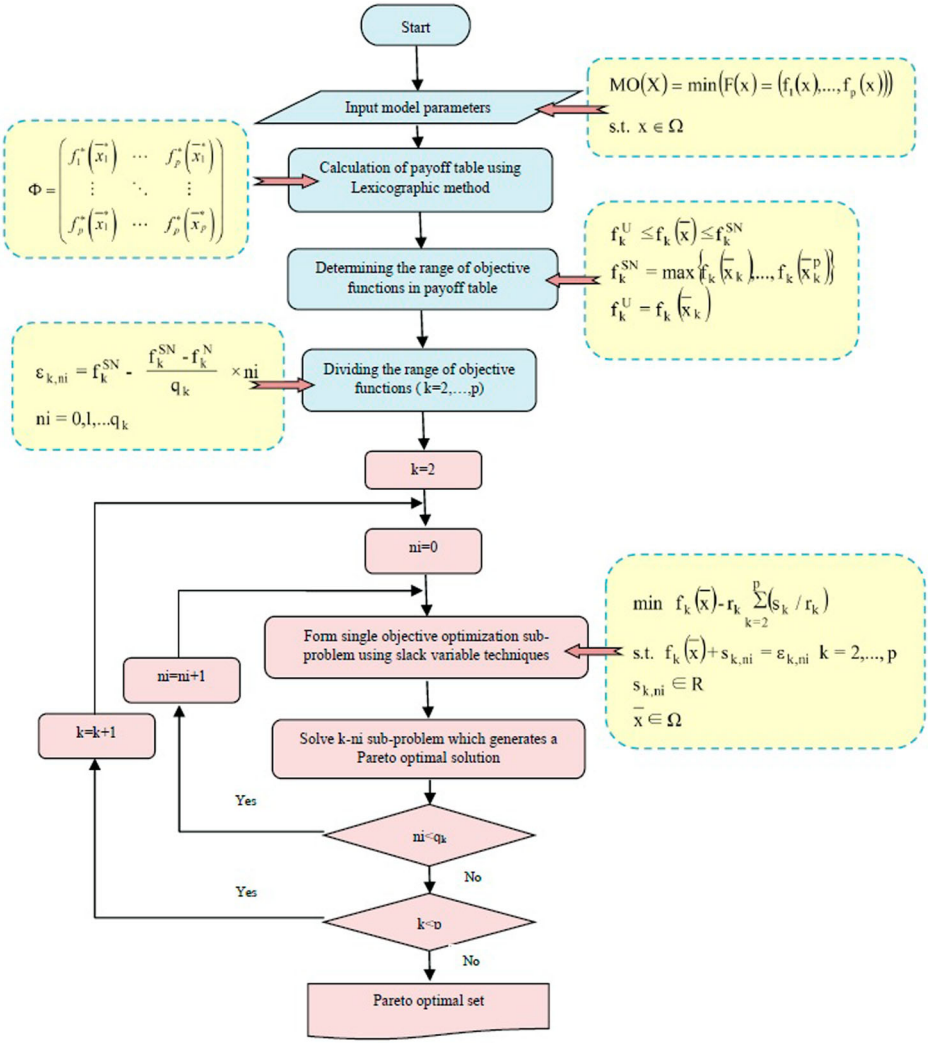


Figure 3. Flowchart of AUGMENCON method for optimization (Adapted from Tartibu, Sun, and Kaunda 2015a, License number: 4520850512009).

6.3. Model implementation

In this section, the two objective components (thermal resistance R_{TH} and pumping power P) are considered simultaneously. Most of the equations involved in the formulation of the multi-objective MPF have been described in Section 4. The optimization problem is formulated as a two-criteria NLP problem that simultaneously minimizes the thermal resistance R_{TH} and pumping power P:

$$(\text{MPF}) \min_{\theta,\phi} \xi = \{R_{TH(\theta,\phi)}; P_{(\theta,\phi)}\} \tag{18}$$

subject to bound limits (12).

In the formulation (18), (θ, ϕ) denotes the parameters describing the geometry of the micro-channel heat sink. There is no single optimal solution that simultaneously optimizes the two objective functions. However, in these cases, the 'most preferred' solutions are sought by the decision makers. To find these solutions, AUGMENCON2, as proposed by Mavrotas and Florios (2013), is

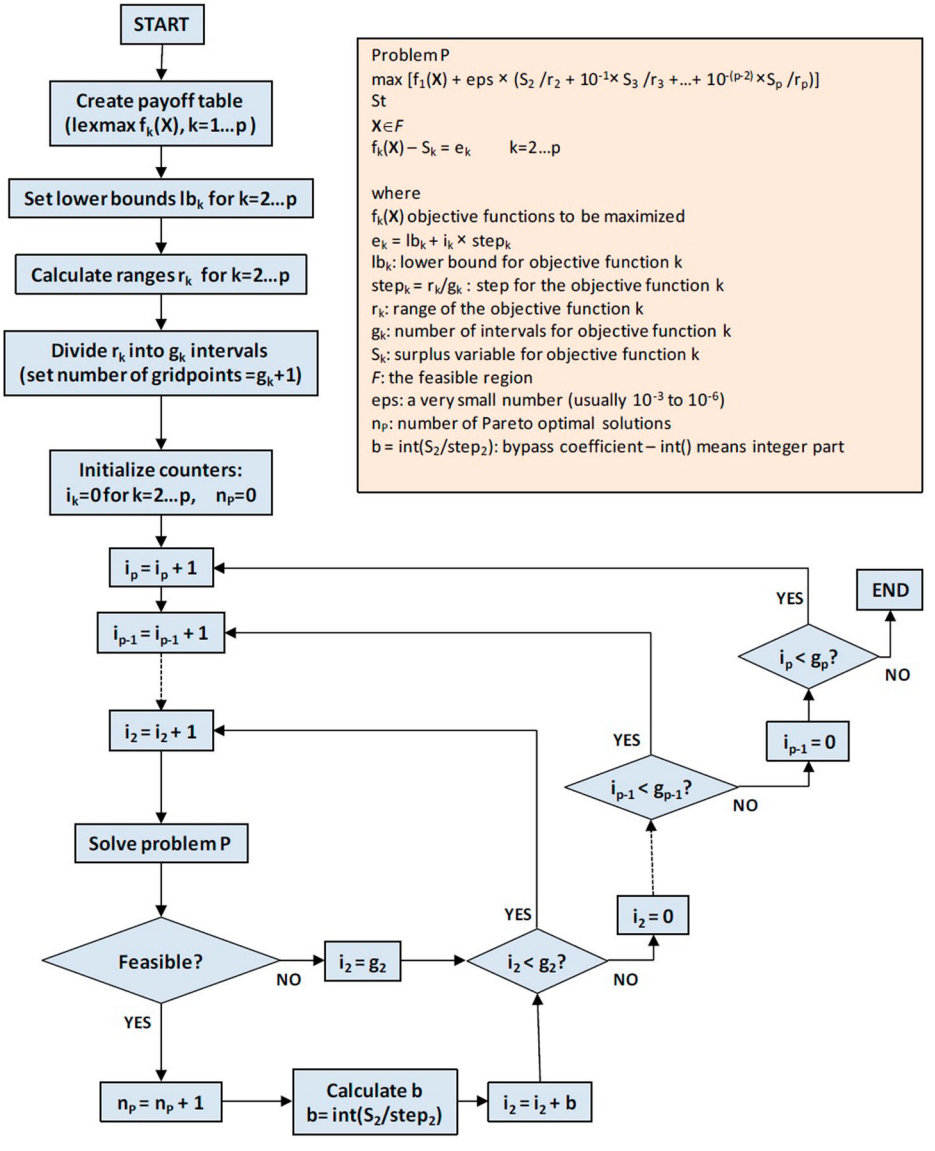


Figure 4. Flowchart of AUGMENCON2 method for optimization (Adapted from Mavrotas and Florios 2013, License number: 4521360993103).

applied. AUGMENCON2 has been coded in GAMS. An example dealing with a multi-objective integer programming problem to illustrate the approach is readily available in the GAMS library (https://www.gams.com/latest/gamslib_ml/libhtml/gamslib_epscmmip.html). Although the parts of the code that perform the calculation of the pay-off table with lexicographic optimization and the generation of Pareto-optimal solutions are fully parameterized and almost ready to use, the part of the code that has to do with the example (specific objective functions, constraints and variables) has been modified to accommodate the micro-channel heat sink problem.

The use of the lexicographic optimization for each objective function to construct the pay-off table for the MPF yields only Pareto-optimal solutions (and avoids weakly efficient solutions) (Mavrotas 2009; Tartibu, Sun, and Kaunda 2015a). The AUGMENCON2 method for solving the model

(Equation 19) can be formulated as follows:

$$\min\left(R_{\mathrm{TH}(\theta,\phi)} + \mathrm{eps} \times \left(\frac{s_2}{r_2}\right)\right) \tag{19}$$

subject to

$$P_{(\theta,\phi)} - s_2 = \varepsilon_2$$

and variable restrictions (12):

$$s_i \in \Re^+$$

where eps is a small number, usually between 10^{-3} and 10^{-6} (Mavrotas and Florios 2013); parameters s_2, r_2 and ε_2 are, respectively, the surplus variables of the respective constraints, the range of objective function ($P_{(\theta,\phi)}$) and the parameters of the right-hand side for the specific iteration drawn from the grid points of the objective functions ($P_{(\theta,\phi)}$). Equation (19) is called the 'augmented ε-constraint method' because of the augmentation of the objective function R_{TH} by the second term. In this method, the lexicographic optimization is performed on each objective function to identify any alternative optima and avoid all weak Pareto-optimal solutions (Mavrotas and Florios 2013). In this study, the range of each objective function is divided into nine intervals (10 grid points). One of the innovative features of AUGMENCON2 is the introduction of the bypass coefficient to avoid redundant iterations and considerably improve the computational speed (Mavrotas and Florios 2013). The maximum central processing unit (CPU) time taken to report the results was 113.444 s in this study.

7. Results and discussion

The AUGMENCON2 algorithm is employed to perform the optimization of the objective functions (Equations 9 and 10). Hence, the Pareto-optimal solutions were computed and the Pareto-optimal front was constructed to analyse the trade-off between thermal resistance and pumping power. The ability of the proposed algorithm was validated by comparing the results obtained with previous works conducted with the NSGA2 in combination with an SQP algorithm (Husain and Kim 2008) and the SPEA2 (Adham, Mohd-Ghazali, and Ahmad 2015). Figure 5 and Table 4 clearly demonstrate that the AUGMENCON2 algorithm is highly capable of obtaining the same results as the NSGA2 combined with SQP algorithm and the SPEA2 algorithm. Therefore, the AUGMENCON2 algorithm has been used to perform the analysis of thermal performance with respect to the geometric configuration of the micro-channel heat sink.

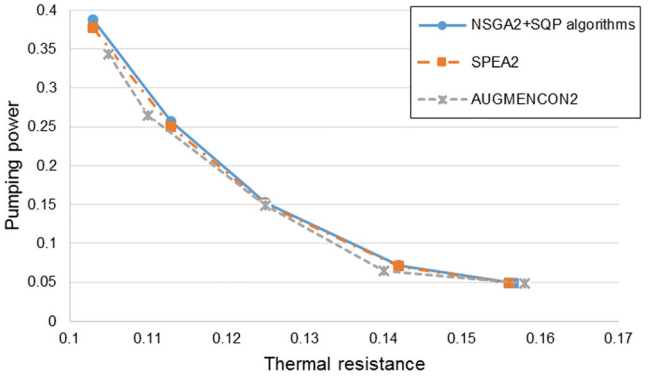


Figure 5. Global Pareto-optimal solutions using the non-dominated sorting genetic algorithm + sequential quadratic programming (NSGA2 + SQP), improved strength Pareto evolutionary algorithm (SPEA2) and improved augmented ε-constraint (AUGMENCON2) method.

Table 4. Pareto-optimal designs obtained using the non-dominated sorting genetic algorithm + sequential quadratic programming (NSGA2 + SQP), improved strength Pareto evolutionary algorithm (SPEA2) and improved augmented ε-constraint method (AUGMENCON2).

	Objective functions values							
	θ	ϕ	NSGA2 + SQP (Husain and Kim 2008b)		SPEA2(Adham, Mohd-Ghazali, and Ahmad 2015)		AUGMENCON2	
Design			R_{TH}	P	R_{TH}	P	R_{TH}	P
A	0.108	0.073	0.103	0.387	0.103	0.377	0.105	0.343
B	0.135	0.074	0.113	0.257	0.113	0.250	0.110	0.264
C	0.164	0.075	0.125	0.152	0.125	0.150	0.125	0.149
D	0.201	0.074	0.142	0.072	0.142	0.070	0.140	0.064
E	0.212	0.047	0.157	0.048	0.156	0.048	0.158	0.048

Table 4 summarizes the results obtained using these approaches.

To obtain greater insight into the Pareto-optimal solutions, the effect of the variation of design variables, namely θ and ϕ, has been analysed. The changes in both thermal resistance and pumping power are plotted against each design variable in Figures 6 and 7.

To compare objective trade-offs among different micro-channel heat sink geometries, value path analysis can be used. An illustration of such an attempt is reported in Devi, Manivannan, and Rao (2012), where the trade-offs among multiple objectives have been determined using value path

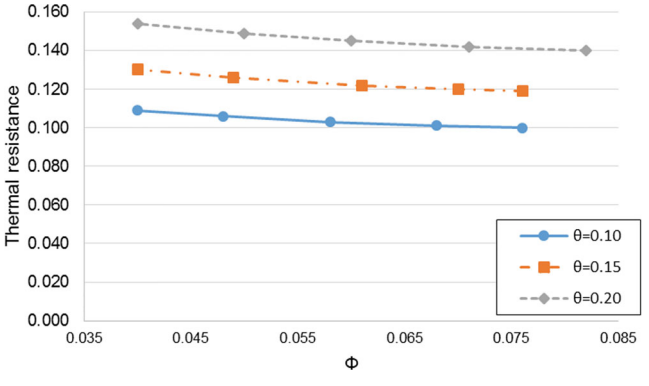


Figure 6. Thermal resistance as a function of design variables.

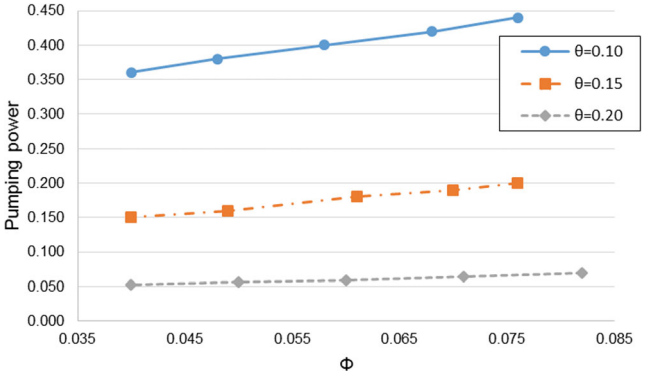


Figure 7. Pumping power as a function of design variables.

Table 5. Summary of value path analysis results for comparison of the non-dominated sorting genetic algorithm + sequential quadratic programming (NSGA2 + SQP), improved strength Pareto evolutionary algorithm (SPEA2) and improved augmented ε-constraint method (AUGMENCON2).

	Objective function values					
	NSGA2 + SQP (Leng *et al.* 2015a)		SPEA2 (Adham, Mohd-Ghazali, and Ahmad 2015)		AUGMENCON2	
Design	R_{TH}	P	R_{TH}	P	R_{TH}	P
A	(1.000) 0.103	(1.128) 0.387	(1.000) 0.103	(1.099) 0.377	(1.019) 0.105	(1.000) 0.343
B	(1.027) 0.113	(1.028) 0.257	(1.027) 0.113	(1.000) 0.250	(1.000) 0.110	(1.056) 0.264
C	(1.000) 0.125	(1.020) 0.152	(1.000) 0.125	(1.007) 0.150	(1.000) 0.125	(1.000) 0.149
D	(1.014) 0.142	(1.125) 0.072	(1.014) 0.142	(1.094) 0.070	(1.000) 0.140	(1.000) 0.064
E	(1.006) 0.157	(1.000) 0.048	(1.000) 0.156	(1.000) 0.048	(1.000) 0.158	(1.000) 0.048

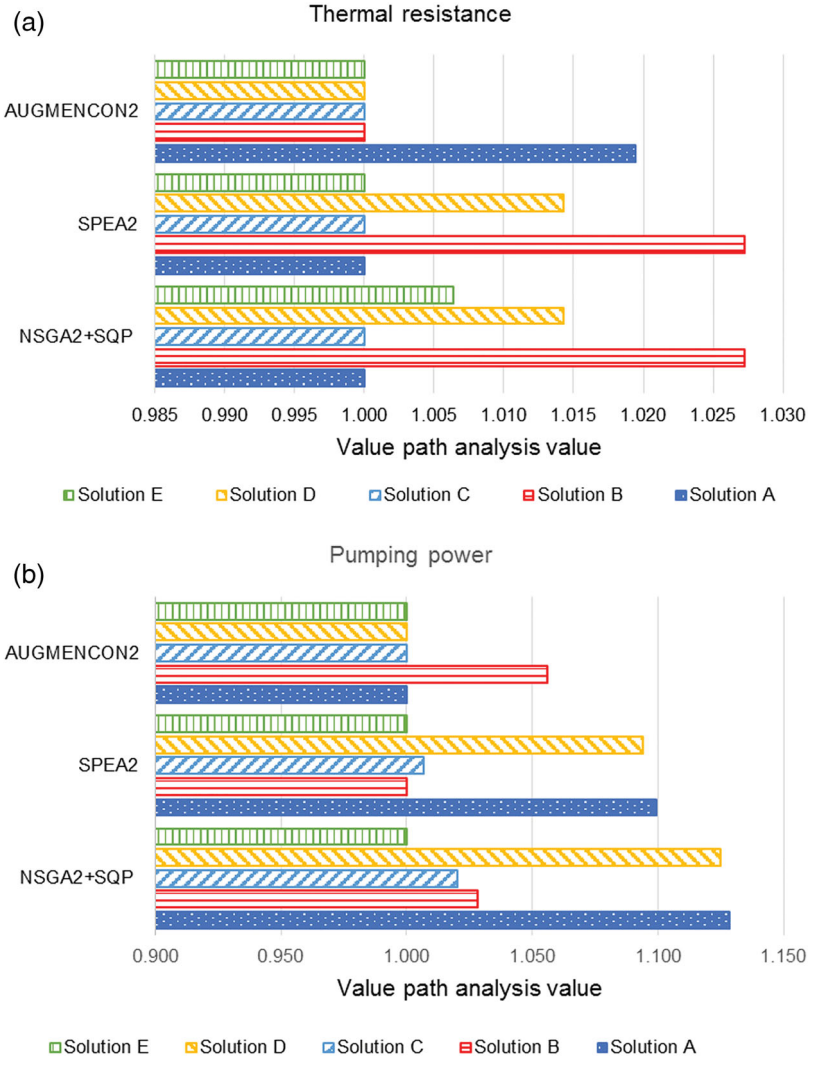


Figure 8. Value path approach for comparison of (a) thermal resistance and (b) pumping power using the improved augmented ε-constraint (AUGMENCON2) method, improved strength Pareto evolutionary algorithm (SPEA2) and non-dominated sorting genetic algorithm + sequential quadratic programming (NSGA2 + SQP).

analysis. This analysis was conducted in steps. First, the best solution achieved for each objective was identified between the optimal solutions (NSGA2 + SQP, SPEA2 and AUGMENCON2), as reported in Table 4. Secondly, the respective objective values of the other objectives were divided by the best value. These results are presented in Table 5, with the value path analysis values for each method reported as a ratio in parenthesis. To compare the different methods used in this study, the results are plotted in Figure 8(a) and (b) showing, respectively, the value path analysis results for comparison of thermal resistance and the value path analysis results for comparison of pumping power. Considering that the decision maker focuses on the minimization of thermal resistance and pumping power, the AUGMENCON2 method proposed in this study clearly achieved better results (Figure 8) than the NSGA2/SQP and the SPEA2 methods, which were randomly selected to validate the results obtained from this study.

8. Conclusion

In this article, a multi-objective optimization approach is discussed that provides a fast initial engineering estimate for the initial design of a rectangular micro-channel heat sink. The thermal performance of a micro-channel is evaluated using two criteria: (1) thermal resistance; and (2) pumping power. Two different parameters, the ratio of micro-channel width and micro-channel depth ($\theta = W_c/H_c$) and the ratio of micro-channel fin width and micro-channel depth ($\phi = W_w/H_c$), describing the geometry of the micro-channel, were investigated. NLP problems have been formulated and implemented in GAMS. For this bi-objective optimization, an improved version of the augmented ε-constraint method (AUGMENCON2) was applied because of its ability to avoid dominated Pareto-optimal solutions. The relationship between thermal performance and the geometric configuration of the micro-channel was elucidated and analysed. The comprehensive design optimization proposed in this study provides clarity about the active design space and the relative sensitivity of design variables describing the geometry of a micro-channel heat sink on thermal performance. This contribution will escalate the efficiency of the handling and manufacturing of design constraints. Moreover, the value path analysis has been used to compare objective trade-offs among various micro-channel heat sink geometries. This analysis suggests that the AUGMENCON2 method proposed in this study performs better than the NSGA2 + SQP and SPEA2 methods, based on the optimal solutions obtained.

Disclosure statement

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References

Adham, A. M., N. Mohd-Ghazali, and R. Ahmad. 2015. "Performance Optimization of a Microchannel Heat Sink Using the Improved Strength Pareto Evolutionary Algorithm (SPEA2)." *Journal of Engineering Thermophysics* 24 (1): 86–100.

Devi, S. P., S. Manivannan, and K. S. Rao. 2012. "Comparison of Nongradient Methods with Hybrid Taguchi-Based Epsilon Constraint Method for Multiobjective Optimization of Cylindrical Fin Heat Sink." *The International Journal of Advanced Manufacturing Technology* 63 (9-12): 1081–1094.

Fedorov, A. G., and R. Viskanta. 2000. "Three-Dimensional Conjugate Heat Transfer in the Microchannel Heat Sink for Electronic Packaging." *International Journal of Heat and Mass Transfer* 43 (3): 399–415.

Harirchian, T., and S. V. Garimella. 2009. "Effects of Channel Dimension, Heat Flux, and Mass Flux on Flow Boiling Regimes in Microchannels." *International Journal of Multiphase Flow* 35 (4): 349–362.

Hetsroni, G., M. Gurevich, A. Mosyak, and R. Rozenblit. 2004. "Drag Reduction and Heat Transfer of Surfactants Flowing in a Capillary Tube." *International Journal of Heat and Mass Transfer* 47 (17): 3797–3809.

Hetsroni, G., A. Mosyak, E. Pogrebnyak, and L. P. Yarin. 2005. "Heat Transfer in Micro-channels: Comparison of Experiments with Theory and Numerical Results." *International Journal of Heat and Mass Transfer* 48 (25): 5580–5601.

Hu, G., and S. Xu. 2009. "Optimization Design of Microchannel Heat Sink Based on SQP Method and Numerical Simulation." In *2009 International Conference on Applied Superconductivity and Electromagnetic Devices*, 89–92. Chengdu: IEEE.

Husain, A., and K. Y. Kim. 2008a. "Multiobjective Optimization of a Microchannel Heat Sink Using Evolutionary Algorithm." *Journal of Heat Transfer* 130 (11): 114505. doi:10.1115/1.2969261

Husain, A., and K. Y. Kim. 2008b. "Optimization of a Microchannel Heat Sink with Temperature Dependent Fluid Properties." *Applied Thermal Engineering* 28 (8): 1101–1107.

Husain, A., and K. Y. Kim. 2008c. "Shape Optimization of Micro-channel Heat Sink for Micro-electronic Cooling." *IEEE Transactions on Components and Packaging Technologies* 31 (2): 322–330.

Husain, A., and K. Y. Kim. 2009. "Microchannel Heat Sinking: Analysis and Optimization." In *Fluid Machinery and Fluid Mechanics*, 185–190. Berlin: Springer.

Husain, A., and K. Y. Kim. 2010. "Enhanced Multi-objective Optimization of a Microchannel Heat Sink Through Evolutionary Algorithm Coupled with Multiple Surrogate Models." *Applied Thermal Engineering* 30 (13): 1683–1691.

Jo, Y. J., C. H. Choe, J. J. Kim, and J. H. Lee. 2001. "Optimum Design of Microchannel Heat Sinks." *Transactions of the Korean Society of Mechanical Engineers B* 25 (1): 117–123.

Karathanassis, I. K., E. Papanicolaou, V. Belessiotis, and G. C. Bergeles. 2013. "Multi-objective Design Optimization of a Micro Heat Sink for Concentrating Photovoltaic/Thermal (CPVT) Systems Using a Genetic Algorithm." *Applied Thermal Engineering* 59 (1): 733–744.

Leng, C., X. D. Wang, T. H. Wang, and W. M. Yan. 2015a. "Multi-parameter Optimization of Flow and Heat Transfer for a Novel Double-Layered Microchannel Heat Sink." *International Journal of Heat and Mass Transfer* 84: 359–369.

Leng, C., X. D. Wang, T. H. Wang, and W. M. Yan. 2015b. "Optimization of Thermal Resistance and Bottom Wall Temperature Uniformity for Double-Layered Microchannel Heat Sink." *Energy Conversion and Management* 93: 141–150.

Li, J., and G. P. Peterson. 2006. "Geometric Optimization of a Micro Heat Sink with Liquid Flow." *IEEE Transactions on Components and Packaging Technologies* 29 (1): 145–154.

Mavrotas, G. 2009. "Effective Implementation of the ϵ-Constraint Method in Multi-objective Mathematical Programming Problems." *Applied Mathematics and Computation* 213 (2): 455–465.

Mavrotas, G., and K. Florios. 2013. "An Improved Version of the Augmented ϵ-Constraint Method (AUGMECON2) for Finding the Exact Pareto Set in Multi-objective Integer Programming Problems." *Applied Mathematics and Computation* 219 (18): 9652–9669.

McHale, J. P., and S. V. Garimella. 2010. "Heat Transfer in Trapezoidal Microchannels of Various Aspect Ratios." *International Journal of Heat and Mass Transfer* 53 (1): 365–375.

Perret, C., C. Schaeffer, and J. Boussey. 1998. "Microchannel Integrated Heat Sinks in Silicon Technology." In *Conference Record of 1998 IEEE Industry Applications Conference. Thirty-Third IAS Annual Meeting (Cat. No.98CH36242)*, 1051–1055. Piscataway, NJ: IEEE.

Rao, R. V., K. C. More, J. Taler, and P. Ocłoń. 2016. "Dimensional Optimization of a Micro-channel Heat Sink Using Jaya Algorithm." *Applied Thermal Engineering* 103: 572–582.

Ryu, J. H., D. H. Choi, and S. J. Kim. 2003. "Three-Dimensional Numerical Optimization of a Manifold Microchannel Heat Sink." *International Journal of Heat and Mass Transfer* 46 (9): 1553–1562.

Tartibu, K., B. Sun, and M. A. E. Kaunda. 2013. "Geometric Optimization of Micro-thermoacoustic Cooler for Heat Management in Electronics." In *Proceedings of the 2013 IEEE International Conference on Industrial Technology (ICIT)*, 527–532. Piscataway, NJ: IEEE.

Tartibu, L. K., B. Sun, and M. A. E. Kaunda. 2015a. "Multi-objective Optimization of the Stack of a Thermoacoustic Engine Using GAMS." *Applied Soft Computing* 28: 30–43.

Tartibu, L. K., B. Sun, and M. A. E. Kaunda. 2015b. "Optimal Design of a Standing Wave Thermoacoustic Refrigerator Using GAMS." *Procedia Computer Science* 62: 611–618.

Tiselj, I., G. Hetsroni, B. Mavko, E. Mosyak, A. Pogrebnyak, and Z. Segal. 2004. "Effect of Axial Conduction on the Heat Transfer in Micro-channels." *International Journal of Heat and Mass Transfer* 47 (12): 2551–2565.

Wang, X. D., B. An, L. Lin, and D. J. Lee. 2013. "Inverse Geometric Optimization for Geometry of Nanofluid-Cooled Microchannel Heat Sink." *Applied Thermal Engineering* 55 (1): 87–94.

Wang, X. D., B. An, and J. L. Xu. 2013. "Optimal Geometric Structure for Nanofluid-Cooled Microchannel Heat Sink Under Various Constraint Conditions." *Energy Conversion and Management* 65: 528–538.

Wang, Z. H., X. D. Wang, W. M. Yan, Y. Y. Duan, D. J. Lee, and J. L. Xu. 2011. "Multi-parameters Optimization for Microchannel Heat Sink Using Inverse Problem Method." *International Journal of Heat and Mass Transfer* 54 (13): 2811–2819.

Xie, X. L., Z. J. Liu, Y. L. He, and W. Q. Tao. 2009. "Numerical Study of Laminar Heat Transfer and Pressure Drop Characteristics in a Water-Cooled Minichannel Heat Sink." *Applied Thermal Engineering* 29 (1): 64–74.